

Military Institute of Science and Technology

Department of Computer Science & Engineering

Undergraduate Thesis

**AI-Enhanced Complaint Management
System for Government Marketplaces**

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Bachelor of Science in Computer Science and Engineering

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Declaration

We hereby declare that this thesis titled *AI-Enhanced Complaint Management System for Government Marketplaces* has been completed as part of the undergraduate research requirement. The work presented in this thesis is based on our own design, implementation, experimentation, and analysis, except where external sources have been cited using standard academic conventions.

We further declare that the dataset, experimental results, and machine learning performance values reported in this thesis are not manually invented. The reported evaluation values are obtained from the implemented training pipeline using the 500-complaint dataset exported from the NityMulya backend. Any limitation related to dataset size, backend-feature dependence, class imbalance, or model generalization has been discussed honestly in the relevant chapters.

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Approval Sheet

This thesis titled *AI-Enhanced Complaint Management System for Government Marketplaces*, submitted by Syed Mafijul Islam, Nahidur Zaman, and Sakif Shahrear, has been accepted as satisfactory in partial fulfillment of the requirements for the degree of Bachelor of Science in Computer Science and Engineering.

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Finally, we thank the users and domain context of the NityMulya marketplace platform, which motivated this work. The research was shaped by a practical goal: to design a lightweight complaint classification pipeline that can support government marketplace administrators without requiring expensive infrastructure or very large language models.

Abstract

Government marketplaces increasingly depend on digital platforms to receive citizen feedback, monitor service quality, and identify irregularities in the sale of essential commodities. In Bangladesh, marketplace complaints are commonly written in Bengali, English, and Banglish, often with informal spelling, mixed scripts, and short descriptions. Manual complaint processing is slow, inconsistent, and difficult to scale. At the same time, many government and public-service systems operate in resource-constrained environments where large transformer models may be costly to train, deploy, and maintain.

This thesis presents a lightweight hybrid Natural Language Processing (NLP) and backend-feature based complaint classification system for the NityMulya government marketplace platform. The study uses a dataset of 500 complaints stored in a Supabase PostgreSQL backend. Each complaint contains text, product category, backend validity score, backend priority score, and a severity label belonging to one of three classes: weak, medium, and high. The proposed pipeline applies Unicode normalization, English and Banglish lowercasing, punctuation cleanup, repeated character normalization, a Banglish normalization dictionary, optional Bengali stopword removal, word-level TF-IDF, character-level TF-IDF, One-Hot Encoding for product category, and scaling for numeric backend features. Logistic Regression, Linear Support Vector Classification, Random Forest, and a Voting ensemble are compared using stratified five-fold cross-validation. Class weighting and SMOTE are also evaluated while avoiding leakage by applying oversampling only inside training folds.

The final selected model is a tuned Logistic Regression classifier using normalized complaint text and backend numeric features. It achieved 86.00% hold-out accuracy and a weighted F1 score of 0.8594 on the 20% stratified test split. The ablation study showed that text-only features achieved only 0.5559 weighted F1, while text combined with backend validity and priority features achieved 0.8503 weighted F1 in cross-validation. Product category features did not improve performance and reduced

weighted F1 in this dataset. These results indicate that the strongest predictive signal came from the fusion of text representation with backend decision-support features rather than from complaint text alone.

The main contribution of this thesis is not a large-scale Bengali language model, but a practical, reproducible, CPU-friendly hybrid pipeline suitable for low-resource government marketplace complaint triage. The thesis also discusses reviewer concerns related to small dataset size, backend features partially encoding operational severity labels, and generalization beyond the NityMulya context.

Keywords: Bengali NLP, Banglish normalization, complaint classification, e-governance, TF-IDF, Logistic Regression, Supabase, backend features, lightweight machine learning.

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Abbreviation List

Abbreviation	Meaning
API	Application Programming Interface
CV	Cross-Validation
DBMS	Database Management System
F1	Harmonic mean of precision and recall
FN	False Negative
FP	False Positive
ICT	Information and Communication Technology
ML	Machine Learning
NFKC	Normalization Form Compatibility Composition
NLP	Natural Language Processing
OHE	One-Hot Encoding
RDBMS	Relational Database Management System
RF	Random Forest
SMOTE	Synthetic Minority Oversampling Technique
SQL	Structured Query Language
SVC	Support Vector Classification
TF-IDF	Term Frequency-Inverse Document Frequency
TN	True Negative
TP	True Positive
UI	User Interface

Chapter 1

Introduction

1.1 Background of E-Governance Systems

E-governance refers to the use of information and communication technologies to improve the delivery, monitoring, and accountability of public services. Modern governments increasingly use digital platforms for citizen registration, subsidy management, online payment, licensing, grievance redress, and marketplace monitoring. In such systems, citizens do not only consume information; they also submit feedback, complaints, documents, and evidence. As a result, the government platform becomes a continuous source of operational data.

Complaint management is one of the most important functions of e-governance because it connects citizens directly with administrative response. A complaint may indicate overpricing, damaged goods, expired products, misleading information, rude shop behavior, or violations of consumer rights. When such complaints are handled quickly, the platform can increase citizen trust and help administrators detect recurring problems. When complaints are delayed or classified inconsistently, citizens may feel that digital governance is merely a form-submission channel rather than a responsive service.

The NityMulya government marketplace platform provides a relevant setting for this research. It is intended to support marketplace information, price awareness, and complaint submission for essential products. Citizens can submit complaints in everyday language rather than in a strictly formal administrative style. This creates both an opportunity and a challenge. The opportunity is that digital complaint records can support data-driven decision-making. The challenge is that complaint text is often short, informal, mixed-language, and noisy.

1.2 Government Marketplace Complaint Management

Government marketplace complaint management differs from general customer-service ticketing. A commercial help desk usually handles issues between a company and its customers. In a government marketplace, however, complaints may involve citizens, shop owners, product categories, legal price ranges, public-health concerns, and administrative enforcement. A complaint about expired milk, false weight, or price manipulation is not only a service problem; it may also require regulatory attention.

The complaint management process normally includes complaint submission, initial review, severity assessment, routing to the responsible authority, investigation, response, and archival. In a manual workflow, administrators must read each complaint and decide whether it is weak, medium, or high priority. This decision can be influenced by the written complaint, the product category, the price context, and backend checks such as whether the charged price exceeded an allowed range. Therefore, a practical classifier should not treat complaint text as the only information source.

1.3 Problems in Manual Complaint Processing

Manual complaint processing creates several operational problems. First, it is slow. If administrators need to read every complaint independently, the review queue grows whenever complaint volume increases. Second, manual classification can be inconsistent. Different officials may interpret similar complaints differently, especially when the text is informal or incomplete. Third, manual processing makes it difficult to identify patterns over time. A digital system can summarize how many complaints are related to expired products or overpricing, but such summaries depend on reliable classification.

Another problem is cognitive fatigue. Repeated reading of short complaint texts can lead to shallow review, especially when many entries are similar. A lightweight classifier cannot replace official judgment, but it can provide a first-level severity signal. Administrators can then focus human attention on high-priority complaints while still retaining oversight.

1.4 Bengali and Banglish NLP Challenges

Bengali NLP introduces challenges that are different from English-only complaint classification. Bengali uses its own script, morphology, and punctuation conventions. Informal digital writing also includes spelling variation and inconsistent spacing. In the NityMulya context, users write complaints in Bengali script, English, and Banglish. Banglish is the use of Bengali words written in Latin characters, often mixed with English words. For example, a user may write “dokan beshi taka niyeche” to mean that a shop charged too much money.

Banglish is difficult because there is no single standardized spelling. The Bengali word for shop may appear as *dokan*, *dokaan*, or mixed with Bengali script. The word for damaged or spoiled may appear as *nosto*, *noshto*, or নষ্টে. A word-level model may treat these as unrelated tokens. Character n-grams and normalization dictionaries can reduce this problem by capturing subword patterns and mapping frequent Romanized terms to Bengali equivalents.

1.5 Motivation of the Study

The motivation of this thesis is practical rather than purely theoretical. Many high-performing NLP systems now use large transformer models such as BERT and its multilingual or Bengali variants [8, 9]. However, such models may require more memory, GPU resources, deployment complexity, and maintenance effort than a small public-service platform can easily support. For a 500-record dataset, a transformer may also overfit unless carefully tuned or supported by additional data.

This thesis therefore investigates whether a classical ML pipeline can produce useful complaint classification performance when text features are combined with backend structured features. The research asks a realistic question: if a government marketplace already stores complaint text, product category, validity score, and priority score, can these fields be fused into a lightweight decision-support classifier? The answer is important for low-resource deployment because TF-IDF and linear models are easier to train, explain, and run on CPU servers.

1.6 Problem Statement

The problem addressed in this thesis is the automatic classification of Bengali, English, and Banglish government marketplace complaints into weak, medium, and high severity classes using a small dataset of 500 complaints. Text-only classification is difficult because complaints are short, noisy, and multilingual. At the same time, backend numeric features such as validity and priority contain structured operational information that can support severity prediction. The research problem is to design, implement, and evaluate a lightweight hybrid classifier that uses both NLP features and backend features without adding new data or reporting invented results.

1.7 Research Objectives

The objectives of this research are:

1. To design a preprocessing pipeline for Bengali, English, and Banglish complaint text.
2. To compare word-level, character-level, and combined TF-IDF representations.
3. To integrate text features with backend product-category and numeric features.
4. To compare Logistic Regression, Linear SVC, Random Forest, and Voting ensemble classifiers.
5. To evaluate class weighting and SMOTE for class imbalance while avoiding data leakage.
6. To perform hyperparameter tuning and stratified five-fold cross-validation.
7. To conduct an ablation study showing the contribution of text, product category, and numeric backend features.
8. To present a thesis-level analysis of results, limitations, and deployment implications.

1.8 Research Questions

This thesis is guided by the following research questions:

1. How effective are classical TF-IDF based models for Bengali, English, and Banglish complaint severity classification?
2. Do character n-grams improve robustness against Banglish spelling variation and mixed-script text?
3. How much do backend validity and priority features contribute compared with text-only features?
4. Which classifier performs best on sparse TF-IDF and structured hybrid features?
5. Does SMOTE improve minority-class performance when applied only inside training folds?
6. What are the deployment advantages and risks of using backend-derived features in a government marketplace classifier?

1.9 Scope of the Study

The scope of the study is limited to severity classification of complaints collected from the NityMulya marketplace platform. The dataset contains 500 records and three target classes: weak, medium, and high. The study does not attempt full legal complaint adjudication, automatic punishment recommendation, named-entity extraction, or multilingual neural translation. It focuses on lightweight classification using classical machine learning.

The work also does not claim that the model fully understands complaint meaning in the same way a human officer does. The strongest model uses backend numeric features, and these features are closely related to the operational rules used in severity assessment. Therefore, the system should be understood as a decision-support component that combines language signals and backend rule-derived signals.

1.10 Research Contribution

The main contributions of this thesis are:

1. A reproducible low-resource complaint classification pipeline for Bengali, English, and Banglish government marketplace complaints.

2. A hybrid feature-fusion design that combines normalized text, product category, validity score, and priority score.
3. A systematic comparison of word TF-IDF, character TF-IDF, combined TF-IDF, Logistic Regression, Linear SVC, Random Forest, Voting ensemble, class weighting, and SMOTE.
4. A clear ablation study demonstrating that backend numeric features are the major contributors to performance, while text alone is insufficient for high accuracy in this dataset.
5. A deployment-oriented discussion explaining why lightweight classical ML can be suitable for CPU-friendly public-service systems.

1.11 Thesis Organization

Chapter 2 reviews literature on NLP, text classification, Bengali and Banglish language processing, TF-IDF, classical classifiers, imbalance handling, and e-governance complaint systems. Chapter 3 describes the NityMulya platform, backend architecture, database design, dataset properties, label definitions, and ethical considerations. Chapter 4 explains the methodology, including preprocessing, feature engineering, model training, cross-validation, tuning, SMOTE, and ablation. Chapter 5 presents implementation details. Chapter 6 reports and analyzes the results. Chapter 7 discusses practical implications and reviewer concerns. Chapter 8 describes limitations and future work. Chapter 9 concludes the thesis. The appendices provide sample data, normalization dictionary entries, search spaces, reports, code snippets, schema examples, and additional visualizations.

Chapter 2

Literature Review

2.1 Introduction to Natural Language Processing

Natural Language Processing is a field of artificial intelligence concerned with computational analysis, representation, and generation of human language. In classification tasks, NLP methods convert raw text into numerical representations that a machine learning model can process. Traditional NLP approaches often rely on tokenization, stopword removal, n-grams, frequency statistics, and classical classifiers. Modern NLP increasingly uses neural embeddings and transformer models, but classical methods remain valuable when data and computational resources are limited.

For complaint classification, the core NLP challenge is to map an unstructured text field to a structured class label. A complaint such as “The shop charged more than the listed price” contains words related to overpricing, but its severity may also depend on price rules and backend validation. Therefore, an NLP classifier for government marketplaces should be evaluated not only as a language model but also as part of a larger decision-support pipeline.

2.2 Text Classification Systems

Text classification assigns one or more labels to a document. Applications include spam detection, sentiment analysis, topic classification, customer support routing, news categorization, and grievance triage. A typical text classification pipeline has four stages: text cleaning, feature extraction, model training, and evaluation. The quality of each stage influences the final result.

In classical text classification, the feature matrix is usually sparse and high-

dimensional. Each column may represent a word, phrase, or character sequence, while each row represents a document. Linear models are commonly strong baselines for such data because they can learn class boundaries directly in sparse vector spaces [2]. Tree-based models may perform well on dense tabular data, but sparse text matrices can be less suitable for axis-aligned splitting.

2.3 Bengali NLP Research

Bengali is one of the most widely spoken languages in the world, yet NLP resources for Bengali remain less mature than for English. Bengali text processing requires script-aware normalization, tokenization, and vocabulary handling. Research has addressed Bengali sentiment analysis, news classification, hate speech detection, and language model pretraining. BanglaBERT and related transformer models show that pretraining on Bengali corpora can improve language understanding tasks [9].

However, many Bengali NLP studies use larger public corpora, social media datasets, or news datasets. Government marketplace complaints are different because they are short, domain-specific, and connected to backend records such as product categories and price validations. This creates a research gap for practical Bengali complaint classification in administrative systems.

2.4 Banglish Language Processing Challenges

Banglish is a code-mixed and transliterated form of Bengali written using Latin characters, often mixed with English and Bengali script. Unlike standard Bengali, Banglish lacks a fixed spelling convention. A user may write “beshi taka”, “beshi tk”, or “bashi product” in the same complaint space. This variation weakens word-level matching and increases vocabulary sparsity.

Approaches to Banglish processing include transliteration, dictionary-based normalization, subword features, and multilingual embeddings. In a small-data setting, a simple dictionary is often easier to audit than a neural transliteration model. Character n-grams are also useful because they represent partial spelling patterns. For example, *dokan*, *dokandar*, and *dokaan* share character sequences that may help a classifier generalize.

2.5 TF-IDF Feature Extraction

Term Frequency-Inverse Document Frequency is a statistical representation that assigns higher weights to terms that are frequent in a document but less common across the corpus [1]. The term frequency component captures local importance, while the inverse document frequency component reduces the influence of very common terms.

For a term t in document d , term frequency may be expressed as:

$$TF(t, d) = \frac{f_{t,d}}{\sum_{t' \in d} f_{t',d}}, \quad (2.1)$$

where $f_{t,d}$ is the frequency of term t in document d . The inverse document frequency can be written as:

$$IDF(t) = \log \left(\frac{N}{1 + df(t)} \right), \quad (2.2)$$

where N is the number of documents and $df(t)$ is the number of documents containing the term. The TF-IDF weight is:

$$TFIDF(t, d) = TF(t, d) \times IDF(t). \quad (2.3)$$

TF-IDF is especially attractive for this thesis because it is transparent, fast, and effective with small datasets. It does not require GPU training or large external corpora.

2.6 Character N-Grams

An n-gram is a sequence of n items. In word n-grams, the items are words. In character n-grams, the items are characters. Character n-grams are useful in noisy multilingual text because they can represent spelling fragments and morphological patterns. The `char_wb` analyzer used in this study creates character n-grams inside word boundaries, reducing noise from cross-word character sequences.

For Bengali and Banglish complaints, character n-grams help with transliteration variation, typographical errors, and mixed scripts. In the experimental comparison, character `char_wb` n-grams with range (3, 5) achieved the strongest TF-IDF variant result under Linear SVC.

2.7 Logistic Regression for NLP

Logistic Regression is a linear classifier that estimates class probabilities from weighted feature sums. For binary classification, the probability of class 1 is:

$$P(y = 1|x) = \sigma(w^T x + b) = \frac{1}{1 + e^{-(w^T x + b)}}. \quad (2.4)$$

For multiclass classification, Logistic Regression extends this idea through one-vs-rest or multinomial formulations. In text classification, Logistic Regression is effective because sparse TF-IDF features often become linearly separable enough for useful decision boundaries. It also provides regularization through the parameter C , where smaller C means stronger regularization.

2.8 Support Vector Machines for Text Classification

Support Vector Machines learn a separating hyperplane that maximizes the margin between classes [3]. Linear SVC is widely used for high-dimensional text features because it focuses on boundary examples and scales well with sparse matrices. The hinge loss objective can be written as:

$$\min_{w,b} \frac{1}{2} \|w\|^2 + C \sum_{i=1}^n \max(0, 1 - y_i(w^T x_i + b)). \quad (2.5)$$

The regularization parameter C controls the trade-off between maximizing margin and reducing classification errors.

2.9 Random Forest for Classification

Random Forest is an ensemble of decision trees trained with bootstrap sampling and random feature selection [4]. It often performs well for tabular data with nonlinear interactions. However, text TF-IDF matrices are sparse and high-dimensional, which can make tree splits less efficient. A Random Forest may need many trees to model useful boundaries, and it may still underperform linear models on sparse text. This expectation was confirmed in the experiments, where Random Forest achieved much lower weighted F1 than Logistic Regression and Linear SVC.

2.10 SMOTE and Imbalance Handling

Class imbalance occurs when one class has many more examples than another. In this dataset, the medium class is the majority class, while weak has fewer records. If imbalance is ignored, a classifier may overpredict the majority class. Class weighting adjusts the loss so minority classes receive higher penalty. SMOTE creates synthetic minority examples by interpolating between minority samples [5]. The synthetic sample can be represented as:

$$x_{new} = x_i + \lambda(x_{nn} - x_i), \quad (2.6)$$

where x_i is a minority sample, x_{nn} is one of its nearest minority neighbors, and $\lambda \in [0, 1]$.

SMOTE must be applied only to training data. If synthetic samples are generated before cross-validation splitting, information from validation folds can leak into training folds. This study avoids that problem by placing SMOTE inside the `imblearn` pipeline.

2.11 E-Governance Complaint Systems

E-governance complaint systems aim to improve administrative responsiveness, transparency, and accountability. A complaint platform can help administrators identify urgent issues, monitor repeated violations, and allocate resources. However, a complaint system must be trustworthy. Automated classification should support human decision-making rather than silently replacing administrative review.

In marketplace regulation, backend data is especially relevant. Product category, expected price range, complaint history, validity score, and priority score can all provide structured evidence. A hybrid approach is therefore more realistic than a text-only model.

2.12 Existing Bengali Complaint Classification Works

Compared with English customer-service classification, Bengali government complaint classification is less represented in published literature. Existing Bengali NLP studies

provide useful methods for representation learning and classification, but most are not designed specifically for government marketplace triage. Table 2.1 summarizes representative work and positions the present thesis.

Table 2.1: Comparison with representative related work

Study	Focus	Method	Relevance and gap
Salton and Buckley [1]	Term weighting	TF-IDF	Establishes the statistical representation used in this thesis.
Joachims [2]	Text categorization	SVM on sparse text	Supports the use of linear models for high-dimensional text data.
Chawla et al. [5]	Imbalanced learning	SMOTE	Provides the oversampling method evaluated in this study.
Devlin et al. [8]	Language modeling	BERT transformer	Shows strength of transformers, but deployment cost is higher.
Bhattacharjee et al. [9]	Bengali language understanding	BanglaBERT	Important Bengali transformer baseline, but not the selected lightweight approach.
This thesis	Bengali/Banglish marketplace complaints	TF-IDF + backend features + classical ML	Focuses on low-resource, deployable hybrid decision support.

2.13 Research Gaps

The literature review identifies four gaps. First, Bengali and Banglish complaint classification for government marketplaces remains underexplored. Second, many NLP studies focus on text alone, while real complaint platforms also store structured backend evidence. Third, transformer models are powerful but may be unnecessary or costly for small datasets and public-service deployment. Fourth, many studies report

performance without detailed ablation analysis, making it difficult to know which feature group actually contributes to the result.

This thesis addresses these gaps through a transparent hybrid pipeline, cross-validation, hyperparameter tuning, imbalance experiments, and an ablation study.

Chapter 3

System Overview and Dataset

3.1 NityMulya Platform Overview

NityMulya is a government marketplace-oriented platform designed to support citizens, shops, and administrators in the context of essential commodity monitoring. The complaint module allows citizens to report issues related to price, product quality, expiry, weight, packaging, and shop behavior. The platform stores complaint records in a Supabase PostgreSQL backend, making the data available for analysis and administrative review.

The system studied in this thesis is not only a text classifier. It is part of a broader marketplace workflow in which complaint text is connected to customer identity, shop identity, product subcategory, and backend rule-based scores. This context is central to the thesis because it explains why backend features improve classification performance.

3.2 System Workflow

The general workflow begins when a citizen submits a complaint through the application. The backend stores the complaint, links it to product and shop information, computes validity and priority scores, and assigns an operational severity class. The machine learning pipeline then exports the relevant fields, preprocesses them, trains models, and evaluates classification performance.

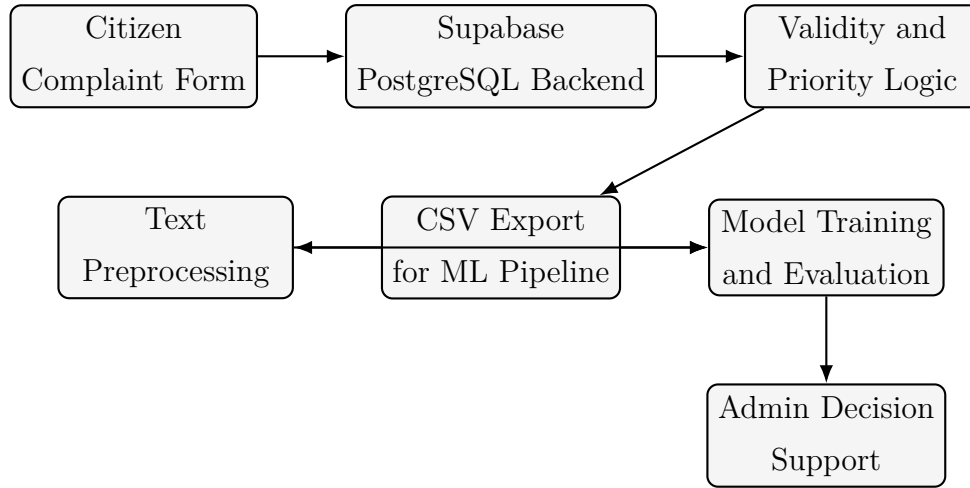


Figure 3.1: High-level workflow of the NityMulya complaint classification system

3.3 Supabase Backend Architecture

Supabase provides a hosted PostgreSQL database with API access, authentication, and storage capabilities. For the complaint module, PostgreSQL acts as the system of record. Complaint entries are stored with identifiers, timestamps, customer information, shop information, product subcategory, complaint text, validity score, priority score, complaint score, and final classification.

The backend architecture is important because the ML model uses features that are available at operational time. Validity and priority are not generated by the ML model; they are backend scores produced by marketplace rules and keyword logic. In deployment, a new complaint would first pass through the backend scoring logic and then through the lightweight classifier.

3.4 PostgreSQL Storage Design

The complaint table stores both unstructured and structured data. The unstructured component is the complaint description. The structured components include product subcategory, validity, priority, and complaint score. The simplified schema is shown in Figure 3.2.

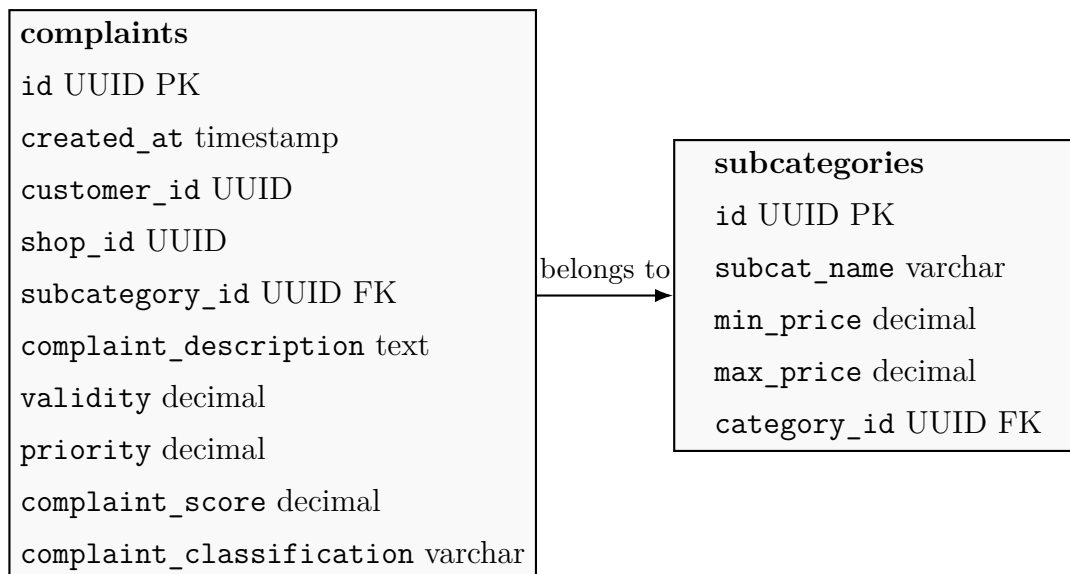


Figure 3.2: Simplified database schema used by the complaint dataset

3.5 Complaint Submission Process

The complaint submission process contains several steps. A citizen selects or enters relevant marketplace information, chooses the product category or subcategory, writes a complaint description, and submits the form. The backend stores the entry and computes rule-based values. Price-related complaints may be compared with product price ranges, while quality or safety complaints may receive higher priority if they contain terms related to expiry, health hazard, fraud, or counterfeit goods.

3.6 Dataset Collection Pipeline

The dataset used in this thesis was exported from the backend into a CSV file named `complaints_full.csv`. The machine learning pipeline loads the CSV, drops records with missing required fields, maps target labels to numeric classes, and separates features from labels. The required fields are complaint description, complaint classification, subcategory name, validity, and priority.

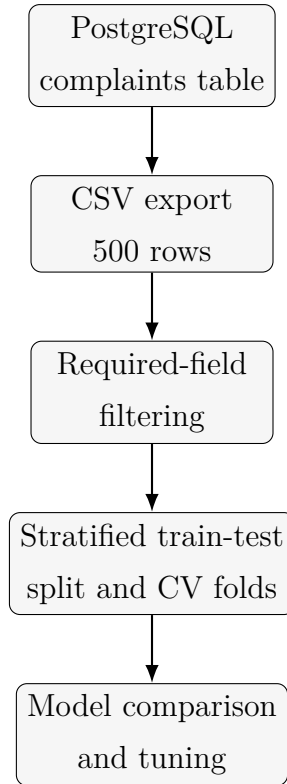


Figure 3.3: Dataset collection and experimental preparation flow

3.7 Dataset Characteristics

The dataset contains 500 complaint entries. The class distribution is imbalanced: 70 weak complaints, 310 medium complaints, and 120 high complaints. The medium class therefore accounts for 62.0% of the dataset. The average validity score is 0.6223, the average priority score is 0.5235, and the average complaint score is 0.5826. Validity ranges from 0.10 to 0.95, priority ranges from 0.20 to 0.95, and complaint score ranges from 0.16 to 0.91.

Table 3.1: Dataset class distribution

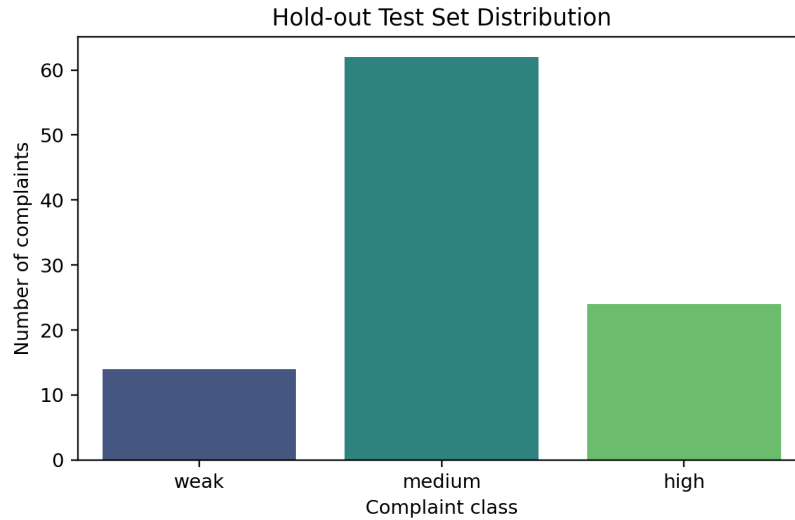
Class	Count	Percentage
Weak	70	14.0%
Medium	310	62.0%
High	120	24.0%

Table 3.2: Backend numeric feature summary

Feature	Minimum	Maximum	Mean
Validity	0.10	0.95	0.6223
Priority	0.20	0.95	0.5235
Complaint score	0.16	0.91	0.5826

3.8 Data Distribution

The stratified hold-out test set contains 100 complaints: 14 weak, 62 medium, and 24 high. This mirrors the overall class distribution and prevents the test split from accidentally overrepresenting or underrepresenting any severity class.

**Figure 3.4:** Distribution of complaint classes in the 20% stratified hold-out test set

3.9 Label Definition

The target label has three classes. Weak complaints represent low-severity or low-validity cases. Medium complaints represent cases that require attention but may not demand immediate action. High complaints represent cases with stronger validity, urgency, or public-risk signals.

The backend documentation defines complaint score as:

$$ComplaintScore = 0.6 \times Validity + 0.4 \times Priority. \quad (3.1)$$

Operational labels are then assigned using score thresholds: high for scores greater than or equal to 0.70, medium for scores from 0.40 to 0.69, and weak for scores below 0.40. This relationship is important for interpretation. Since validity and priority help define the operational label, they naturally carry strong predictive information. Using them is appropriate for a backend decision-support model, but the result should not be interpreted as evidence that text alone solves the task.

3.10 Complaint Examples

Table 3.3 presents representative examples showing Bengali, English, and Banglish variation. Personal identifiers are omitted.

Table 3.3: Representative complaint examples

Language	Complaint text	Class
Bengali	সয়াবিন তেল এর গুণমান খারাপ ছিল। পচা এবং দুর্গন্ধযুক্ত।	High
English	The product quality was very poor. Rotten and smelly.	High
Banglish	Dokane expired dim bikri kore. Swasther jonno khotikar.	High
Mixed	Bashi morich bikri koreche. Taka ferot chay nai.	Medium

3.11 Ethical Considerations

Complaint data can contain personal information, shop names, and allegations. A complaint classification system must therefore be designed with privacy and fairness in mind. This thesis uses exported research data for model evaluation and does not require disclosure of personally identifiable information in the thesis text. Examples are limited and should be anonymized in any public version.

The classifier should not be used as the sole authority for legal or punitive decisions. It should support prioritization and routing, while human administrators retain responsibility for final decisions. This is particularly important when model predictions are wrong or when a complaint contains unusual details not captured by the training data.

3.12 Data Privacy Discussion

Data privacy requires minimizing unnecessary personal data, restricting access to raw complaints, and logging model outputs responsibly. If deployed, the system should store predictions with timestamps and model version identifiers so that decisions can be audited. Sensitive fields such as customer identifiers and contact details should not be used in the ML feature set unless there is a clear legal and ethical basis. This thesis uses complaint text, product subcategory, validity, and priority only.

Chapter 4

Methodology

4.1 Overall Machine Learning Pipeline

The methodology follows a structured pipeline: data loading, preprocessing, feature extraction, feature fusion, model training, cross-validation, hyperparameter tuning, final hold-out evaluation, and result reporting. Figure 4.1 summarizes the pipeline.

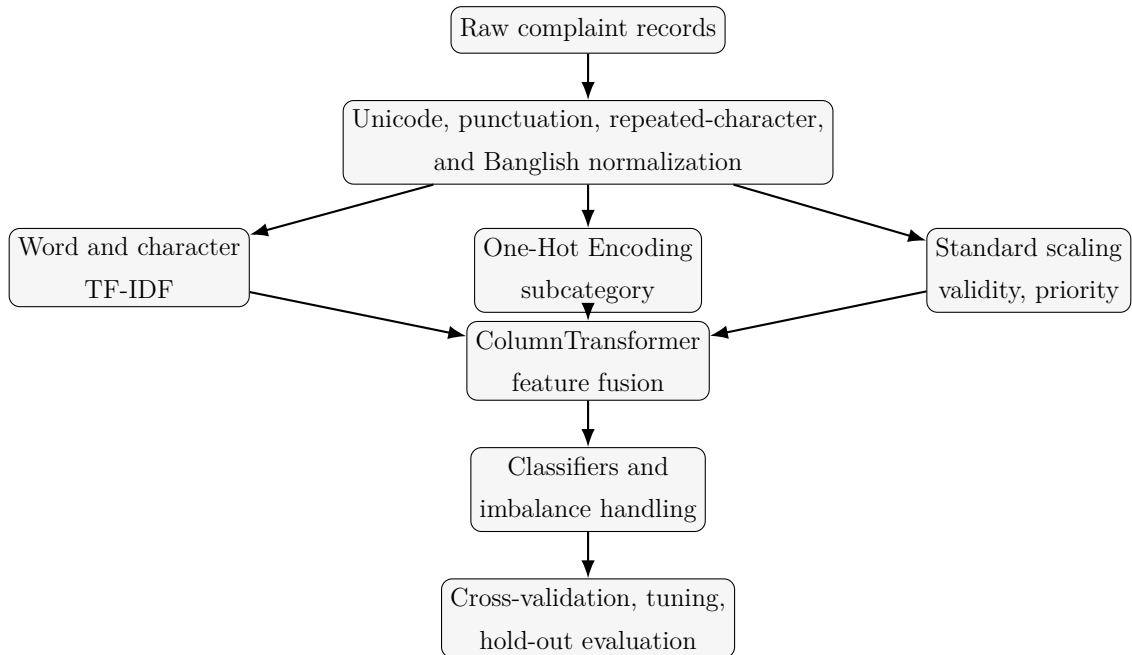


Figure 4.1: Machine learning pipeline used in the thesis

4.2 Text Preprocessing Pipeline

The preprocessing pipeline is designed for Bengali, English, and Banglish complaint text. The steps are:

1. Convert missing values to empty strings.
2. Apply Unicode NFKC normalization.
3. Lowercase English and Banglish tokens.
4. Compress repeated characters.
5. Remove noisy punctuation while preserving Bengali, Latin letters, digits, and spaces.
6. Replace frequent Banglish marketplace terms using a dictionary.
7. Optionally remove Bengali stopwords.

The goal is not to over-clean the text. Excessive preprocessing can remove useful complaint signals. Instead, the pipeline targets common noise while preserving domain words such as product names, price terms, expiry terms, and quality terms.

4.3 Unicode Normalization

Unicode normalization converts visually similar character sequences into a consistent representation. This study uses NFKC normalization. In multilingual text, different input methods can create compatibility characters or inconsistent forms. Normalization reduces unnecessary variation before tokenization and vectorization.

4.4 Banglish Normalization

Banglish normalization maps frequent Romanized marketplace terms to Bengali equivalents. Examples include *dokan* to দোকান, *dam* to দাম, *taka* to টাকা, *chal* to চাল, and *nosto* to নষ্ট. This dictionary is intentionally simple and auditable. A large neural transliteration system could be explored in future work, but the current thesis prioritizes transparency and low-resource deployment.

4.5 Stopword Removal

Stopwords are frequent function words that may carry less class-specific information. Bengali stopwords removal was implemented as an optional parameter and selected during hyperparameter tuning for the final Logistic Regression model. However, stopwords removal is not always beneficial in short texts, because even common words can contribute to phrase patterns. Therefore, it was treated as a tunable choice rather than a fixed assumption.

4.6 TF-IDF Mathematics

TF-IDF transforms text into numeric vectors by combining local term frequency with corpus-level rarity. In this thesis, both word-level and character-level TF-IDF features are evaluated. Sublinear term frequency is used so that repeated words do not dominate linearly. If tf is the raw term frequency, sublinear scaling uses:

$$tf_{sublinear} = 1 + \log(tf). \quad (4.1)$$

The final TF-IDF matrix is sparse because each complaint contains only a small subset of possible words or character n-grams. Sparse matrices are memory-efficient and work well with linear classifiers.

4.7 Word N-Gram Extraction

Word n-grams represent sequences of words. A unigram model uses individual words, while a bigram model includes adjacent two-word phrases. For example, “expired product” and “beshi taka” can be more informative than the words alone. This study compares word n-gram ranges $(1, 1)$ and $(1, 2)$.

4.8 Character N-Gram Extraction

Character n-grams represent subword sequences. The character `char_wb` n-gram range $(3, 5)$ is used because three-to-five-character fragments can capture meaningful spelling patterns without producing excessively long sequences. Character n-grams are particularly helpful for Banglish spelling variation and incomplete words.

4.9 FeatureUnion Explanation

When word and character TF-IDF are combined, the pipeline uses a **FeatureUnion**. A **FeatureUnion** runs multiple feature extractors in parallel and concatenates their outputs. If X_w is the word TF-IDF matrix and X_c is the character TF-IDF matrix, the combined text matrix is:

$$X_{text} = [X_w \ X_c]. \quad (4.2)$$

This allows the model to use both complete-token meaning and subword robustness.

4.10 Numeric Feature Scaling

The backend numeric features are validity and priority. These values are scaled using **StandardScaler** with sparse compatibility. Scaling is useful because models such as Logistic Regression and Linear SVC are sensitive to feature magnitudes. If μ is the mean and σ is the standard deviation, scaling is:

$$z = \frac{x - \mu}{\sigma}. \quad (4.3)$$

4.11 One-Hot Encoding

Product category or subcategory is categorical. One-Hot Encoding converts each category into a binary feature. Unknown categories are ignored during transformation so that the model can process categories that were not seen during training. However, the ablation study showed that product category did not improve this dataset and reduced weighted F1 when added to text.

4.12 Cross-Validation Methodology

Stratified five-fold cross-validation was used to estimate model performance. Stratification preserves class proportions in each fold. For each fold, the model trains on four parts and validates on the remaining part. If the fold metric values are $m_1, m_2, \dots, m_5$, the mean and standard deviation are:

$$\bar{m} = \frac{1}{5} \sum_{i=1}^5 m_i, \quad (4.4)$$

$$s = \sqrt{\frac{1}{5} \sum_{i=1}^5 (m_i - \bar{m})^2}. \quad (4.5)$$

4.13 Hyperparameter Tuning

RandomizedSearchCV was used to tune candidate models. Randomized search samples combinations from a defined parameter space and evaluates them using cross-validation. It is more efficient than exhaustive grid search when several hyperparameters exist. The tuned parameters included TF-IDF maximum features, word n-gram range, optional Bengali stopwords removal, Logistic Regression C , Linear SVC C , and Random Forest tree parameters.

4.14 Class Imbalance Handling

The dataset is imbalanced, with the medium class being the majority class. Logistic Regression and Linear SVC used `class_weight='balanced'`, which increases the loss contribution of minority classes. SMOTE was also tested. To avoid leakage, SMOTE was placed inside the training pipeline so that synthetic samples were created only from the training portion of each fold.

4.15 Ablation Study Methodology

An ablation study removes or adds feature groups to measure their contribution. Four feature sets were evaluated:

1. Text only.
2. Text plus product category.
3. Text plus backend numeric features.
4. Full hybrid feature set: text, product category, and numeric features.

The ablation study is central to this thesis because it reveals that backend numeric features are the main reason performance improves.

4.16 Why Linear Models Outperform Random Forest on Sparse TF-IDF

Sparse TF-IDF vectors have many dimensions and many zero values. Linear models can assign weights directly to useful terms and learn a separating hyperplane. Random Forest models split features one at a time and may struggle when useful evidence is distributed across many sparse dimensions. In small datasets, tree ensembles may also overfit local patterns or fail to find stable splits. This explains why Linear SVC and Logistic Regression outperformed Random Forest in the experiments.

4.17 Pseudocode

Listing 4.1: Pseudocode for the experimental pipeline

```
Input: Complaint dataset D with text, subcategory, validity, priority,
      label
Output: Evaluation tables, final classifier, confusion matrix

1. Load D from complaints_full.csv
2. Drop rows with missing required fields
3. Map labels: weak -> 0, medium -> 1, high -> 2
4. Create stratified 80/20 train-test split
5. For each TF-IDF variant:
    preprocess text
    extract features
    evaluate Linear SVC with 5-fold StratifiedKfold
6. For each classifier and imbalance setting:
    build hybrid feature pipeline
    optionally apply SMOTE inside training folds
    evaluate with 5-fold StratifiedKfold
7. Tune Logistic Regression, Linear SVC, and Random Forest
8. Select best tuned model family by weighted F1
9. Run ablation study using best tuned settings
10. Fit final model on training split
11. Evaluate final model on hold-out test split
12. Save reports, tables, and visualizations
```


Chapter 5

Implementation

5.1 Development Environment

The implementation was developed in Python using a CPU-friendly machine learning stack. The main training script is located at `ML_Model_Training/src/pipeline.py`. The dataset is stored at `ML_Model_Training/data/complaints_full.csv`, and the generated metrics and figures are stored in `ML_Model_Training/results/`.

5.2 Python Ecosystem

Python was selected because it provides strong libraries for data processing, machine learning, evaluation, and visualization. The main libraries are Scikit-learn, Imbalanced-learn, NumPy, Matplotlib, Seaborn, and Pandas. Scikit-learn provides the pipeline, vectorizer, classifier, and cross-validation tools. Imbalanced-learn provides SMOTE and pipeline support for leakage-safe oversampling.

5.3 Scikit-Learn

Scikit-learn is used for `TfidfVectorizer`, `ColumnTransformer`, `FeatureUnion`, `LogisticRegression`, `LinearSVC`, `RandomForestClassifier`, `VotingClassifier`, `StratifiedKFold`, `cross_validate`, and `RandomizedSearchCV`. The use of pipelines is important because preprocessing, feature extraction, scaling, oversampling, and classification are treated as a single reproducible object.

5.4 Imbalanced-Learn

Imbalanced-learn provides the pipeline that can include SMOTE between feature extraction and classification. This is necessary because Scikit-learn's standard pipeline does not treat oversampling as a typical transformer. By placing SMOTE inside the cross-validation workflow, the implementation prevents validation data from influencing synthetic training samples.

5.5 Supabase Integration

Supabase PostgreSQL stores complaint data. The exported CSV is used for offline model training and evaluation. In a deployment setting, the trained model can be loaded by a backend service, and the service can classify new complaints after the backend computes validity and priority. The Supabase table remains the source of operational data, while the ML model adds decision-support predictions.

5.6 Data Loading

The data-loading function reads the CSV file, checks required columns, removes incomplete rows, maps class labels to integers, and returns feature and target matrices. The relevant feature columns are complaint description, subcategory name, validity, and priority.

Listing 5.1: Data-loading structure from the training pipeline

```
required_columns = [
    "complaint_description",
    "complaint_classification",
    "subcategory_name",
    "validity",
    "priority",
]
df = df.dropna(subset=required_columns).copy()
df["target"] = df["complaint_classification"].map(CLASS_MAPPING)
X = df[["complaint_description", "subcategory_name", "validity", "priority"]
      ]
y = df["target"]
```

5.7 Pipeline Implementation

The feature transformer uses `ColumnTransformer`. The text column is processed by the normalizer and TF-IDF pipeline. The product subcategory is processed by One-Hot Encoding when included. Validity and priority are scaled when numeric features are included.

Listing 5.2: Simplified hybrid feature transformer

```
transformers = [  
    ("text", build_text_pipeline(...), "complaint_description")  
]  
  
if feature_set in {"text_category", "full"}:  
    transformers.append(  
        ("category", OneHotEncoder(handle_unknown="ignore"), ["  
            subcategory_name"])  
    )  
  
if feature_set in {"text_numeric", "full"}:  
    transformers.append(  
        ("numeric", StandardScaler(with_mean=False), ["validity", "priority  
        "])  
    )
```

5.8 Model Training Implementation

Each model is wrapped in the same pipeline interface. This ensures that all classifiers receive comparable preprocessing and feature representations. Logistic Regression and Linear SVC use balanced class weights. Random Forest is tested with class weighting and tree hyperparameters.

5.9 Evaluation Implementation

The evaluation reports accuracy, weighted precision, weighted recall, and weighted F1. Weighted metrics are appropriate because the dataset is imbalanced and each class should contribute according to its support. The final hold-out report also includes

class-wise precision, recall, and F1.

5.10 Visualization Generation

The pipeline generates a class distribution plot, confusion matrix, two-dimensional SVD projection of transformed features, and K-Means cluster projection. These figures are used in Chapter 6 to support interpretation. The SVD plots are exploratory visualizations, not primary evidence of classifier performance.

5.11 Training Workflow Diagram

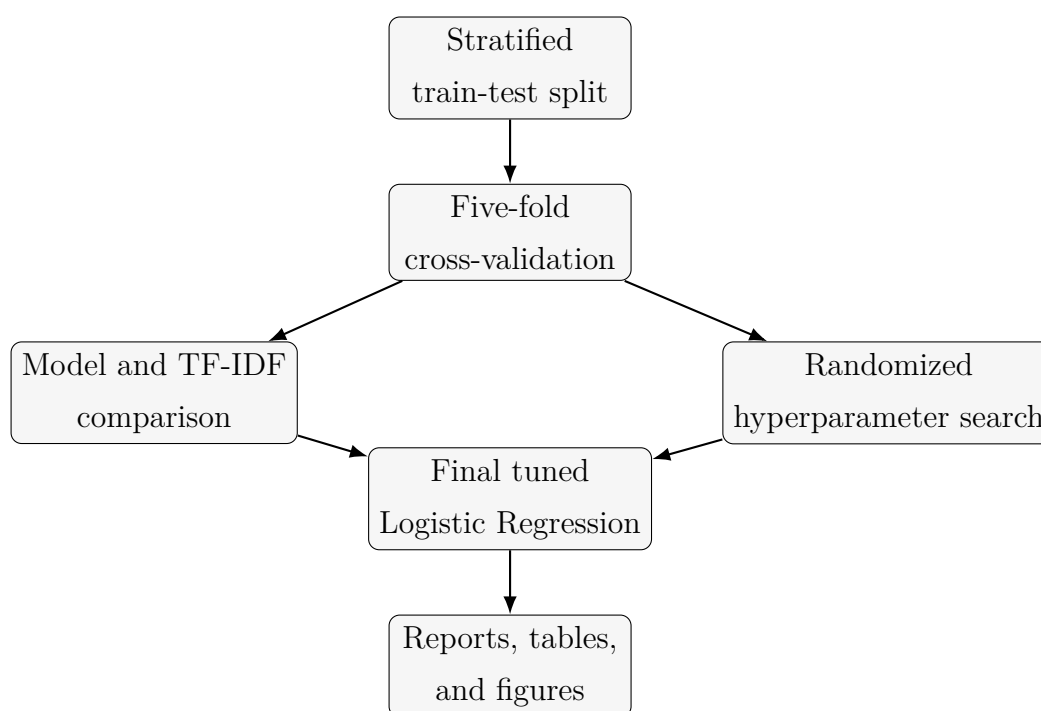


Figure 5.1: Training and evaluation workflow

5.12 Deployment Possibilities

The trained model can be deployed as a backend service in several ways. It can run as a Python microservice called from the Node.js or application backend. It can also be serialized using `joblib` and loaded into a scheduled batch job that classifies new complaints. Because the selected model is Logistic Regression with TF-IDF and two numeric features, inference is lightweight and CPU-friendly. This makes it suitable for modest cloud instances and government systems where GPU access is limited.

Chapter 6

Results and Analysis

6.1 Evaluation Metrics

The study uses accuracy, weighted precision, weighted recall, and weighted F1. Precision measures how many predicted examples of a class are correct. Recall measures how many actual examples of a class are found. F1 is the harmonic mean of precision and recall:

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}. \quad (6.1)$$

Weighted F1 averages class F1 scores by class support, making it suitable for imbalanced datasets.

6.2 Baseline Random Forest Results

The original Random Forest baseline achieved 58.00% accuracy and 0.58 weighted F1 on the same 20% hold-out test split. The weak class F1 was approximately 0.30. This result showed that the initial model was not sufficient for reliable triage, especially for minority-class behavior.

After improving preprocessing, feature extraction, model comparison, imbalance handling, tuning, and ablation, the best final model achieved much stronger performance. It is important to note that this improvement came primarily from model and feature design, not from adding more data.

6.3 Improved Logistic Regression Results

The final selected model was tuned Logistic Regression using normalized text and backend numeric features. It achieved 86.00% hold-out accuracy and 0.8594 weighted F1. Table 6.1 reports class-wise performance.

Table 6.1: Final hold-out classification report for tuned Logistic Regression

Class	Precision	Recall	F1-score	Support
Weak	0.9231	0.8571	0.8889	14
Medium	0.8750	0.9032	0.8889	62
High	0.7826	0.7500	0.7660	24
Macro average	0.8602	0.8368	0.8479	100
Weighted average	0.8596	0.8600	0.8594	100

The weak and medium classes achieved strong F1 scores, while the high class remained more difficult. The high class had 24 examples in the test split, and six high complaints were misclassified as medium. This suggests that the boundary between medium and high remains a practical challenge.

6.4 TF-IDF Comparison Analysis

Table 6.2 presents the five-fold TF-IDF comparison using Linear SVC.

Table 6.2: Five-fold TF-IDF comparison using Linear SVC

Feature extractor	Accuracy	Weighted F1	F1 std.
Character <code>char_wb</code> (3,5)	0.8220	0.8225	0.0286
Word unigram + bigram	0.8220	0.8209	0.0359
Word + character	0.8100	0.8092	0.0342
Word unigram	0.8040	0.8041	0.0432

Character n-grams achieved the highest weighted F1 among TF-IDF variants. This supports the expectation that subword features help with spelling variation, Banglish text, and mixed-script complaints. Word unigrams and bigrams also performed

competitively, showing that phrase-level signals such as “expired product” or “beshi taka” are useful.

6.5 Model Comparison Analysis

Table 6.3 reports the classifier and imbalance comparison on the full hybrid feature set.

Table 6.3: Five-fold model comparison on the full hybrid feature set

Classifier	Imbalance handling	Accuracy	Weighted F1	F1 std.
Linear SVC	Class weight	0.8100	0.8092	0.0342
Voting Ensemble	Class weight	0.8020	0.8013	0.0314
Logistic Regression	Class weight	0.7980	0.8000	0.0291
Linear SVC	Class weight + SMOTE	0.7820	0.7814	0.0378
Voting Ensemble	Class weight + SMOTE	0.7580	0.7601	0.0320
Logistic Regression	Class weight + SMOTE	0.7400	0.7461	0.0344
Random Forest	Class weight	0.6460	0.6224	0.0492
Random Forest	Class weight + SMOTE	0.6160	0.6083	0.0309

Linear SVC achieved the best default cross-validation score. Logistic Regression was selected after tuning because it produced the strongest tuned family and the best final hold-out behavior with the selected feature set. Random Forest performed poorly compared with linear models, which is consistent with the sparse TF-IDF explanation in Chapter 4.

6.6 Cross-Validation Analysis

Cross-validation gives a more reliable estimate than a single split because every record is used for validation once. The best default row, Linear SVC without SMOTE, achieved 0.8092 weighted F1 with a standard deviation of 0.0342. The tuned Logistic Regression family achieved 0.8093 weighted F1 during tuning and 0.8249 full-hybrid cross-validation weighted F1 when evaluated using the tuned configuration.

The difference between cross-validation and hold-out performance should be interpreted cautiously. The hold-out test set has only 100 examples, so the exact value can vary depending on the split. The more important conclusion is the consistent ranking: linear models perform better than Random Forest, and numeric backend features greatly improve performance.

6.7 SMOTE Analysis

SMOTE did not improve the results. In every classifier family, SMOTE reduced weighted F1 compared with class weighting alone. This may be because synthetic interpolation in sparse high-dimensional feature space creates artificial vectors that do not correspond to realistic complaint patterns. For small text datasets, class weighting is often safer than oversampling. The implementation still tested SMOTE correctly by applying it only within training folds.

6.8 Ablation Study Analysis

The ablation study is the most important result for understanding the system. Table 6.4 shows that text alone achieved only 0.5559 weighted F1. Adding product category reduced performance slightly to 0.5410. Adding numeric backend features increased weighted F1 to 0.8503. The full feature set achieved 0.8249, lower than text plus numeric features.

Table 6.4: Ablation study with tuned Logistic Regression

Feature set	Accuracy	Weighted F1	F1 std.
Text + numeric features	0.8500	0.8503	0.0299
Full hybrid feature set	0.8260	0.8249	0.0247
Text only	0.5580	0.5559	0.0419
Text + product category	0.5400	0.5410	0.0328

This result clearly shows that complaint text alone did not achieve high performance. The text contains useful signals, but many complaints are short, repetitive, and semantically overlapping. For example, both medium and high complaints may mention poor quality or expired products. The backend validity and priority scores provide structured information that separates severity levels more effectively.

6.9 Why Backend Numeric Features Improved Results

Validity and priority improved results because they represent domain knowledge already computed by the backend. Validity captures whether the complaint appears legitimate according to price or complaint rules. Priority captures urgency, health risk, fraud terms, or quality concern. These scores reduce ambiguity in text. A short complaint saying “bad product” may be difficult to classify from text alone, but if the backend priority is high and validity is high, the model can assign a more appropriate severity.

At the same time, this finding must be interpreted carefully. Since validity and priority contribute to the operational complaint score used to define labels, they are not independent evidence in the same way raw text is. They are legitimate deployment-time features, but they also mean the model partly learns the backend severity policy. This is acceptable for a decision-support system whose goal is consistent operational triage, but it limits claims about pure NLP understanding.

6.10 Why Product Category Features Reduced Performance

Product category features reduced performance in the ablation study. There are several possible reasons. First, the dataset has many product subcategories compared with only 500 records, so category features may be sparse and unstable. Second, severity may depend less on the product type and more on validity and urgency. Third, some product categories appear in all severity classes, making category a weak predictor. Fourth, adding category features may introduce noise that competes with stronger numeric features.

6.11 Confusion Matrix Analysis

Figure 6.1 shows the final confusion matrix. The model correctly classified 12 of 14 weak complaints, 56 of 62 medium complaints, and 18 of 24 high complaints. Most errors occurred between medium and high classes.

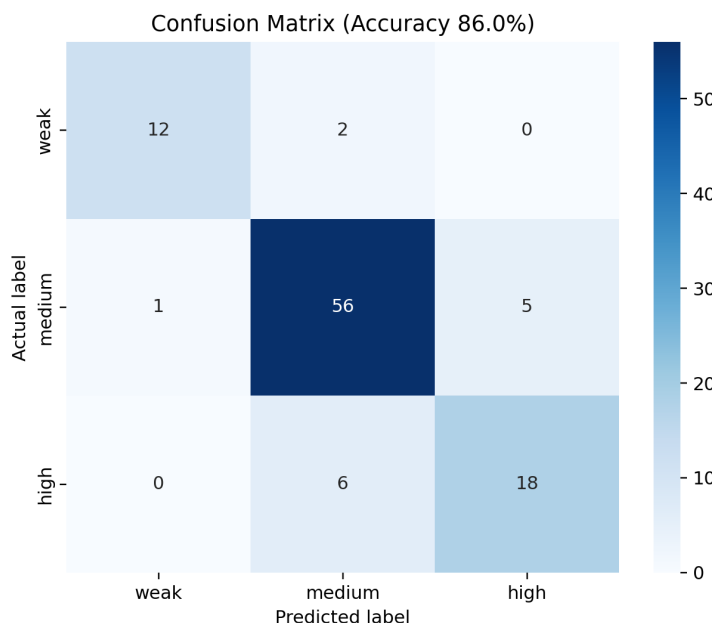


Figure 6.1: Confusion matrix for the final tuned Logistic Regression model

The confusion matrix suggests that the model rarely confuses weak complaints with high complaints. This is encouraging for triage because the most serious error would be assigning a high-severity complaint to weak. In the hold-out result, no high complaint was predicted as weak. However, six high complaints were predicted as medium, indicating that medium-high separation should be improved in future work.

6.12 Feature Importance Discussion

For linear models, feature weights can provide interpretability. Positive weights for a class indicate that a feature increases the model's confidence for that class. In this system, backend numeric features are expected to have strong weights because they correlate with severity. Text features related to expiry, health risk, fake products, poor quality, overpricing, and weight manipulation may also contribute.

Unlike Random Forest feature importance, linear model coefficients are more natural for sparse TF-IDF. Future work could export class-wise top weighted terms to help administrators understand model decisions.

6.13 Error Analysis

Errors mainly occur where complaint severity is ambiguous. A complaint may use strong language but have low backend validity. Another complaint may have valid pricing evidence but mild wording. Some Banglish complaints may contain uncommon spellings not covered by the dictionary. Product names in Bengali may also vary in spelling or formatting.

The high class is especially important. A high complaint predicted as medium still receives attention, but perhaps not with the desired urgency. The absence of high-to-weak errors in the test split is a useful sign, but the dataset is small and this should be verified on larger future data.

6.14 Class-Wise Analysis

The weak class improved substantially compared with the original baseline. The final weak F1 score is 0.8889 on the hold-out test set. This improvement is likely because backend validity helps identify low-severity or invalid complaints. The medium class also achieved 0.8889 F1 because it has the largest number of examples and occupies the central score range. The high class achieved 0.7660 F1, which is acceptable but weaker than the other classes.

6.15 Statistical Interpretation

The standard deviations in cross-validation indicate that results are moderately stable but still affected by the small dataset. A five-fold standard deviation of about 0.03 in weighted F1 means that fold composition matters. Therefore, the reported 86.00% hold-out accuracy should not be treated as a guaranteed production value. It should be interpreted together with cross-validation, ablation, and domain analysis.

6.16 SVD and Cluster Visualization

Figure 6.2 shows a two-dimensional SVD projection of the transformed feature space. Figure 6.3 shows K-Means clusters over the same reduced feature components. These plots help visualize whether classes or clusters show separable structure, but they should not be overinterpreted because high-dimensional data loses information when

projected to two dimensions.

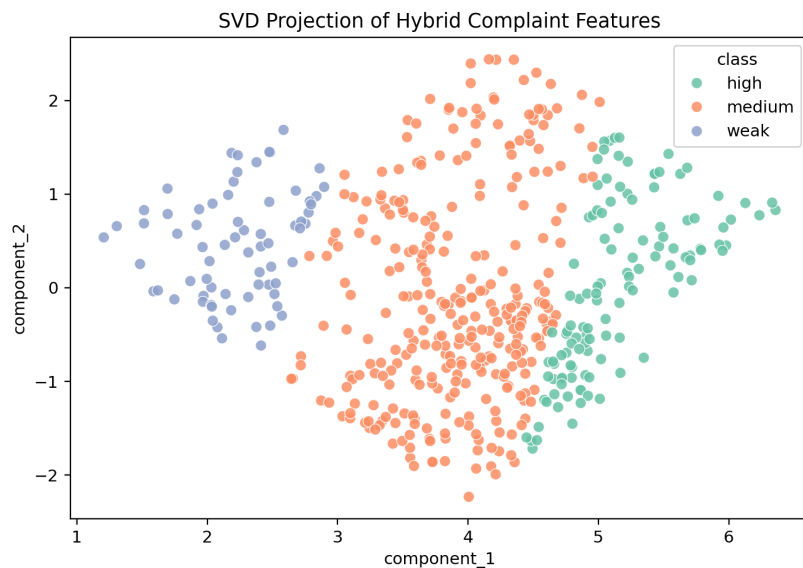


Figure 6.2: Two-dimensional SVD projection of hybrid complaint features

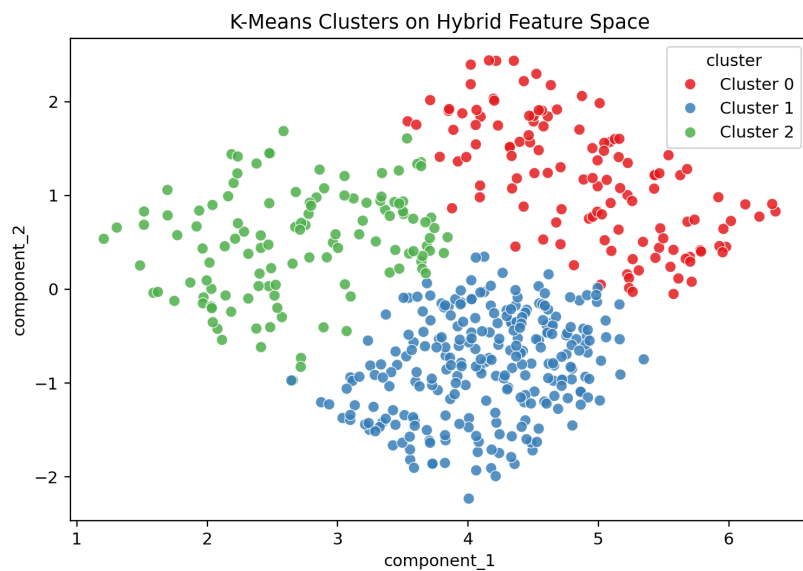


Figure 6.3: K-Means cluster projection over transformed hybrid features

Chapter 7

Discussion

7.1 Interpretation of Results

The results show that a lightweight hybrid model can support complaint severity classification in a Bengali government marketplace context. The strongest evidence comes from the ablation study. Text-only classification was weak, while text plus backend numeric features achieved high weighted F1. Therefore, the system should be described as a hybrid NLP and backend-data classifier, not as a pure NLP solution.

7.2 Why the Hybrid System Works

The hybrid system works because it combines complementary information. Text features capture what the citizen wrote. Character n-grams handle spelling variation and Banglish. Backend numeric features capture operational knowledge that may not be fully expressed in text. A complaint may be short, vague, or emotionally written, but backend validity and priority can provide structured severity evidence.

7.3 Practical Deployment Implications

In deployment, the model can provide an initial severity prediction as soon as a complaint is submitted and backend scores are computed. Administrators can sort complaint queues by predicted severity, review high-severity cases first, and monitor recurring patterns. The model can also be retrained periodically as new labeled complaints become available.

7.4 Lightweight ML Advantages

The selected approach has several advantages. It trains quickly, runs on CPU, requires limited memory, and can be inspected more easily than a large transformer. It is also easier to reproduce in a university or government setting. For small datasets, classical models can be more reliable than over-parameterized neural models.

7.5 CPU-Friendly Deployment Discussion

TF-IDF vectorization and Logistic Regression inference are computationally inexpensive. A trained vectorizer and classifier can process many complaints per second on ordinary hardware. This matters for government systems where infrastructure budgets, GPU availability, and maintenance capacity may be limited.

7.6 Risks of Backend-Feature Dependency

The main risk is that backend features may encode the label too directly. In this dataset, validity and priority contribute to complaint score, and complaint score defines severity thresholds. Therefore, strong performance partly reflects the model’s ability to learn the backend scoring policy. This is not leakage if the same features are available at prediction time, but it limits the claim of independent semantic classification.

To address this risk, the thesis reports text-only results separately. The text-only result makes clear that language features alone did not produce high performance. Future work should include human-reviewed labels independent of backend score thresholds to test whether the model generalizes beyond the current policy.

7.7 Generalization Concerns

The dataset contains only 500 complaints from one platform context. It may not represent all regions, product types, complaint styles, or administrative policies. Banglish spellings may also vary widely across users. Generalization should therefore be evaluated on future complaints collected after deployment.

7.8 Real-World Applicability

Despite limitations, the system is applicable as a triage assistant. It can reduce manual workload by ordering complaints and highlighting likely high-priority cases. It should not replace investigation, enforcement, or final administrative judgment. The best use case is first-level decision support.

7.9 Comparison with Transformer-Based Systems

Transformer models such as BERT and BanglaBERT can capture deeper context and may outperform TF-IDF when enough labeled data and compute are available [8,9]. However, they are more expensive to train and deploy. For this thesis, the lightweight model is a deliberate design choice. It provides strong performance through feature fusion without requiring transformer fine-tuning.

7.10 Reviewer Concern: Data Leakage

Data leakage occurs when information unavailable at prediction time influences training or validation. This study avoids common ML leakage by using stratified splits, applying preprocessing inside pipelines, and applying SMOTE only within training folds. However, backend-feature dependence is a separate concern. Validity and priority are legitimate prediction-time features, but they are related to label creation. This is discussed transparently and treated as a limitation of independent NLP evaluation.

7.11 Reviewer Concern: Backend Features Encoding Labels

Because complaint labels are generated from complaint score thresholds, and complaint score is calculated from validity and priority, the model can learn a simplified version of the backend scoring rule. This is defensible only if the research objective is operational decision support. It would not be defensible to claim that the model learned complaint severity from language alone. For this reason, the thesis repeatedly distinguishes between text-only performance and hybrid performance.

7.12 Reviewer Concern: Small Dataset Size

A 500-record dataset is small for NLP, especially for three-class classification. The study mitigates this through cross-validation, simple models, regularization, class weighting, and ablation analysis. Still, larger datasets are necessary before production claims can be made. The thesis positions the work as a practical undergraduate prototype and baseline for future expansion.

Chapter 8

Limitations and Future Work

8.1 Small Dataset Limitations

The dataset size is the most important limitation. With only 500 records, rare complaint patterns may not be represented. The test split contains only 100 records, so a small number of errors can change the final percentage substantially. Future studies should collect more data over a longer period.

8.2 Class Imbalance

The weak class has fewer examples than the medium class. Although class weighting improved minority-class handling, imbalance remains a concern. Additional weak and high examples should be collected to make the classifier more robust.

8.3 Backend-Rule Dependency

The model depends strongly on backend validity and priority scores. This is useful for operational consistency but limits independent language understanding. If backend rules change, the model may need retraining. If backend scores contain bias or errors, the model may reproduce them.

8.4 Banglish Normalization Limitations

The Banglish dictionary is simple and cannot cover all transliteration variants. Users may write the same Bengali word in many Romanized forms. Future work should evaluate automatic transliteration, phonetic matching, or subword neural models.

8.5 Lack of Transformer Benchmarking

This thesis intentionally focuses on lightweight classical ML. It does not fine-tune BanglaBERT or multilingual BERT on the dataset. A transformer benchmark would be useful for understanding the performance-cost trade-off more completely.

8.6 Generalization Limitations

The dataset comes from a specific marketplace setting. The classifier may not generalize to other government services, legal complaint portals, or private e-commerce platforms without retraining. External validation is needed.

8.7 Future Work

Future work should include:

1. Collecting a larger and more diverse complaint dataset.
2. Creating human-reviewed labels independent of backend score thresholds.
3. Benchmarking BanglaBERT, multilingual BERT, and other transformer models.
4. Using active learning so uncertain complaints are prioritized for human labeling.
5. Investigating online learning for periodic model updates.
6. Developing a human-in-the-loop dashboard for reviewer feedback.
7. Adding model explainability reports with top weighted terms and numeric contributions.
8. Integrating the classifier into the production mobile and backend workflow.

9. Monitoring fairness, false negatives, and drift after deployment.

Chapter 9

Conclusion

9.1 Summary of the Work

This thesis presented a lightweight hybrid NLP and backend-feature based complaint classification system for Bengali government marketplaces. The system was developed using 500 NityMulya complaint records containing Bengali, English, and Banglish text, product subcategory, validity score, priority score, and severity label. The pipeline applied text normalization, Banglish dictionary mapping, TF-IDF feature extraction, feature fusion, class weighting, SMOTE comparison, hyperparameter tuning, cross-validation, ablation, and final hold-out evaluation.

9.2 Summary of Findings

The final tuned Logistic Regression model achieved 86.00% hold-out accuracy and 0.8594 weighted F1. The ablation study showed that text-only features achieved only 0.5559 weighted F1, while text plus backend numeric features achieved 0.8503 weighted F1. This finding confirms that backend validity and priority features were the major contributors to performance. Product category features did not improve results in this dataset.

9.3 Contributions

The thesis contributes a reproducible, low-resource, deployment-oriented complaint classification pipeline. It demonstrates that classical machine learning can be useful when combined with structured backend features. It also provides a careful analysis

of limitations, especially the fact that backend numeric features are related to the operational label definition.

9.4 Practical Significance

For a government marketplace platform, the proposed system can support faster complaint triage, reduce manual workload, and help administrators identify high-priority cases. Because the model is lightweight, it can run on CPU infrastructure and can be integrated into backend services without the cost of transformer deployment.

9.5 Final Remarks

The thesis shows that practical AI for public-service systems does not always require the largest model. In a low-resource setting, careful preprocessing, feature engineering, backend-data fusion, and honest evaluation can produce a useful decision-support system. The next step is to collect more data, add independent human labels, compare transformer baselines, and evaluate the system in a real administrative workflow.

Appendix A

Sample Dataset Entries

Table A.1: Sample anonymized complaint records

Subcategory	Complaint text	Validity	Class
ডিম (ফার্ম)	Dokane expired ডিম bikri kore. Swasther jonno khotikar.	0.64	High
এ্যাংকর ডাল	এ্যাংকর ডাল kinte giye protoronar shikar hoyechi. Fake product diyeche.	0.65	Medium
আটা (প্যাকেট)	Dokane expired আটা bikri kore. Swasther jonno khotikar.	0.30	Medium
সয়াবিন তেল	সয়াবিন তেল এর গুণমান খারাপ ছিল। পচা এবং দুর্গন্ধযুক্ত।	0.72	High

Appendix B

Banglish Normalization Dictionary

Table B.1: Selected Banglish normalization entries

Banglish token	Normalized form
alu	আলু
bashi	বাসি
beshi	বেশি
chal	চাল
dam	দাম
dal	ডাল
dim	ডিম
dokan	দোকান
dokandar	দোকানদার
dudh	দুধ
expired	মেয়াদোত্তীর্ণ
fake	নকল
kharap	খারাপ
kom	কম
nosto	নষ্ট
oil	তেল
price	দাম
receipt	রসিদ

rice	চাল
taka	টাকা

Appendix C

Hyperparameter Search Space

Table C.1: Hyperparameter search space

Parameter	Candidate values
Bengali stopwords removal	False, True
Word n-gram range	(1,1), (1,2)
Word TF-IDF max features	500, 1000, 2000
Character TF-IDF max features	500, 1000, 2000
SMOTE	passthrough, SMOTE(k=3)
Logistic Regression C	0.1, 0.5, 1.0, 2.0, 5.0, 10.0
Linear SVC C	0.1, 0.5, 1.0, 2.0, 5.0, 10.0
Random Forest n_estimators	100, 200, 300
Random Forest max_depth	None, 8, 12, 20
Random Forest min_samples_split	2, 5, 10

Appendix D

Full Classification Reports

Table D.1: Confusion matrix rows=true, columns=predicted

	Weak	Medium	High
Weak	12	2	0
Medium	1	56	5
High	0	6	18

Table D.2: Hyperparameter tuning results

Classifier	Best mean weighted F1	Std.
Logistic Regression	0.8093	0.0409
Linear SVC	0.8045	0.0407
Random Forest	0.6547	0.0559

Appendix E

Source Code Snippets

Listing E.1: Text normalization logic

```
text = unicodedata.normalize("NFKC", text)
text = text.lower()
text = LATIN_REPEAT_RE.sub(r"\1\1", text)
text = BENGALI_REPEAT_RE.sub(r"\1\1", text)
text = PUNCTUATION_RE.sub(" ", text)

tokens = []
for token in text.split():
    normalized = BANGGLISH_MAP.get(token, token)
    if remove_bengali_stopwords and normalized in BENGALI_STOPWORDS:
        continue
    tokens.append(normalized)
```

Listing E.2: SMOTE placed inside the training pipeline

```
smote_step = "passthrough"
if use_smote:
    smote_step = SMOTE(k_neighbors=3, random_state=RANDOM_STATE)

pipeline = ImbPipeline(
    steps=[
        ("features", build_feature_transformer(...)),
        ("smote", smote_step),
        ("classifier", build_classifier(model_key)),
    ]
)
```


Appendix F

Supabase Schema Examples

Listing F.1: Simplified complaint table schema

```
CREATE TABLE complaints (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP,  
  customer_id UUID NOT NULL,  
  shop_id UUID NOT NULL,  
  subcategory_id UUID REFERENCES subcategories(id),  
  complaint_description TEXT NOT NULL,  
  validity DECIMAL(3,2),  
  priority DECIMAL(3,2),  
  complaint_score DECIMAL(3,2),  
  complaint_classification VARCHAR(20)  
);
```

Appendix G

Additional Visualizations

Additional generated visualizations are stored in the thesis folder under **figures/**. These include the hold-out class distribution, final confusion matrix, SVD feature-space projection, and K-Means projection. In a final university submission, screenshots of the NityMulya complaint form, administrator dashboard, and training console may also be added in this appendix.

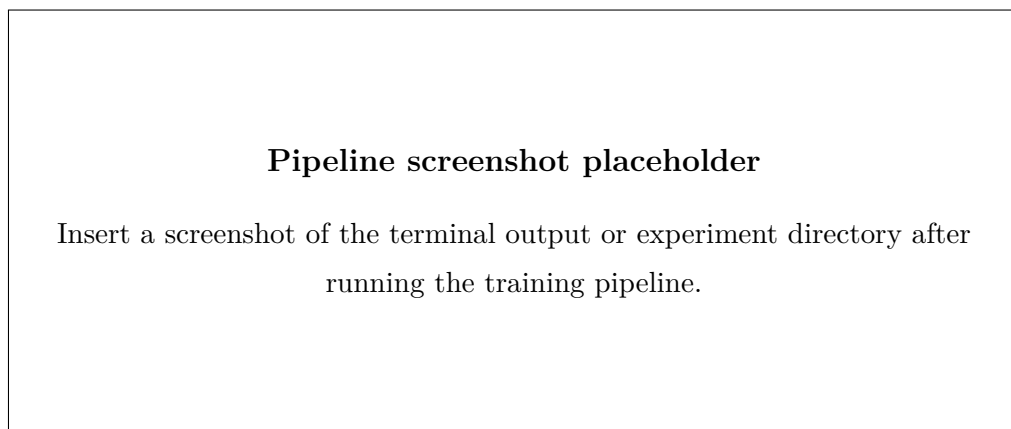


Figure G.1: Pipeline screenshot placeholder

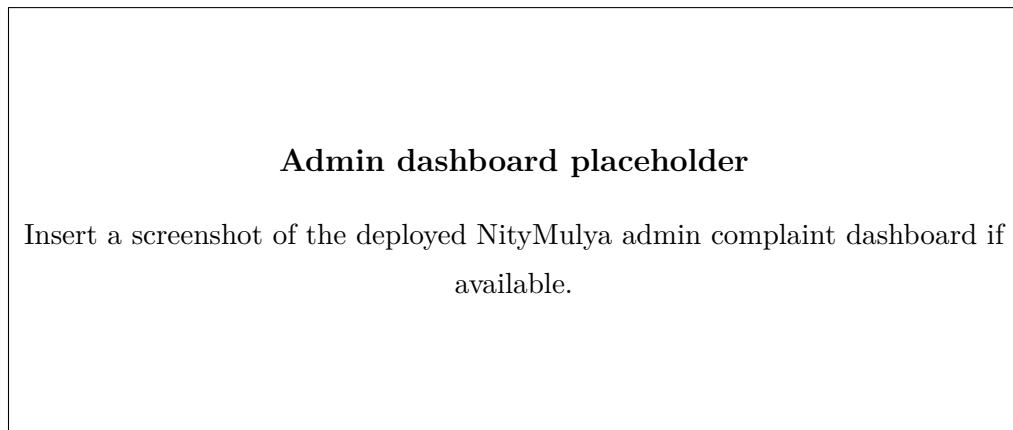


Figure G.2: Admin dashboard placeholder

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